BE700 Homework 3: Insulin Resistance, Diabetes and Aging in Human Skeletal Muscle

Jinjie Liu

October 22, 2026

Contents

1 Introduction and hypotheses **1**

2 Data and preprocessing **2**
2.1 Quality control 4
2.2 Outlier samples 6

3 Heatmap of the muscle transcriptome **7**

4 Clustering patients **9**
4.1 Are clusters enriched for phenotypes? 11

5 Genes correlated with age **12**
5.1 Sensitivity of age associations to flagged samples 14
5.2 Heatmap of the exploratory age-ranked features 15
5.3 Correlation plots for the top age-correlated genes 17

6 Principal component analysis **18**

7 Additional analysis: insulin sensitivity and diabetes **24**

8 DAVID functional enrichment (Part III) **26**

9 Discussion **30**

10 AI usage and reproducibility **31**

11 References **31**

1 Introduction and hypotheses

Insulin resistance in skeletal muscle precedes type 2 diabetes (T2D), and aging is itself a major risk factor for both. Jin et al. (2011) profiled vastus lateralis biopsies from normoglycemic people with or without a family history (FH) of T2D and from people with T2D, and reported that increased serum response factor (SRF) activity marks insulin resistance. The Hallmarks of Aging (López-Otín et al., 2013) predicts that age-associated changes should involve mitochondrial dysfunction, deregulated nutrient sensing, altered intercellular communication (inflammation), and loss of proteostasis.

Hypotheses tested here:

- **H1.** Unsupervised clustering of muscle transcriptomes separates samples by a biological factor (sex, diabetes status, or age), not at random.
- **H2.** Sex is the strongest single source of transcriptome-wide variation, driven by sex-chromosome genes.
- **H3.** A set of genes is correlated with age, and these genes are enriched for Hallmark-of-Aging processes (mitochondria / oxidative phosphorylation, extracellular matrix, inflammation).
- **H4.** Because the diabetic group is older than the other groups, part of any apparent "age signature" is confounded by diabetes; this must be tested explicitly.

2 Data and preprocessing

- **Expression:** the course-provided `BE700_human_expression_1.csv` for GSE25462. The assignment explicitly states that this matrix is already log-normalized; no second log transform or additional normalization is applied here.
- **Phenotypes:** `BE700_human_pheno_1.csv`, containing the GEO sample metadata and the phenotype columns ending in `:ch1`.
- **Provenance boundary:** the metadata identifies GPL570 (Human Genome U133 Plus 2.0) and records GCOS v1.3/MAS5.0 processing with a trimmed-mean scaling target of 1500. This upstream processing is distinct from the log-transformed, annotated course matrix. The exact probe-to-feature aggregation script and annotation version were not supplied. The paper's main text describes U133A 2.0 and groups of 15/26/11 people, whereas GEO and the supplied matrix contain 15/25/10. This analysis uses the 50 supplied samples and their GPL570 metadata, not the paper's larger demographic totals.
- **Unit of analysis:** rows are gene-annotated expression features. Entries such as `A///B` have ambiguous multi-gene annotations; splitting them for annotation does not establish that the expression of A and B was measured independently.

```
expr_raw <- read.csv("data/BE700_human_expression_1.csv", check.names = FALSE,
                     stringsAsFactors = FALSE)
genes <- expr_raw[[1]]
keep  <- !is.na(genes) & genes != ""
expr  <- as.matrix(expr_raw[keep, -1]); rownames(expr) <- genes[keep]

ph_raw <- read.csv("data/BE700_human_pheno_1.csv", check.names = FALSE,
                   stringsAsFactors = FALSE)
num <- function(x) suppressWarnings(as.numeric(x))   # "missing value" -> NA
ph <- data.frame(
  sample   = ph_raw$geo_accession,
  title    = ph_raw$title,
  sex      = factor(ph_raw[["gender:ch1"]]),
  age      = num(ph_raw[["age (years):ch1"]]),
  bmi      = num(ph_raw[["body mass index (kg/m2):ch1"]]),
  whr      = num(ph_raw[["waist hip ratio:ch1"]]),
  hba1c    = num(ph_raw[["hemoglobin a1c:ch1"]]),
  fglu     = num(ph_raw[["fasting glucose (iv0gavg):ch1"]]),
  finsulin = num(ph_raw[["fasting insulin (iv0inavg):ch1"]]),
  si       = num(ph_raw[["si:ch1"]]),
```

```
  glu2h    = num(ph_raw[["2 hour glucose (ogtt):ch1"]]),
  stringsAsFactors = FALSE)
ph$group <- factor(sub("^Muscle-", "", sub("-[0-9]+$", "", ph$title)),
                   levels = c("FamilyHistoryNegative", "FamilyHistoryPositive",
                   ↪  "Diabetes"),
                   labels = c("FH-", "FH+", "T2D"))
ph$label    <- paste0(c("FH-" = "N", "FH+" = "P", "T2D" =
↪  "D")[as.character(ph$group)],
                      sub(".*-", "", ph$title))      # short plot labels, e.g. D3,
                      ↪  N12
ph$diabetes <- factor(ph$group == "T2D", labels = c("No", "Yes"))
rownames(ph) <- ph$sample

stopifnot(identical(colnames(expr), ph$sample),
          !anyDuplicated(ph$sample), !anyDuplicated(rownames(expr)),
          is.numeric(expr), all(is.finite(expr)),
          all(apply(expr, 1, var) > 0),
          !anyNA(ph[, c("age", "sex", "group", "diabetes")]))
n_multimapping <- sum(grepl("///", rownames(expr), fixed = TRUE))
dim(expr)
```

```
## [1] 22189    50
```

The matrix has **22189 expression features x 50 samples** after removing 1 row with an empty annotation. Sample order matches the phenotype table; expression values are finite and row identifiers are unique. Of the retained rows, 1341 contain /// multi-gene annotations. Gene names below refer to these annotations, with this measurement ambiguity retained.

```
summary_vars <- c("age", "bmi", "hba1c", "si")
cohort_summary <- do.call(rbind, lapply(levels(ph$group), function(g) {
  rows <- ph$group == g
  out <- data.frame(group = g, n = sum(rows),
                    female = sum(ph$sex[rows] == "female"))
  for (v in summary_vars) {
    x <- ph[[v]][rows]
    out[[paste0(v, "_n")]] <- sum(!is.na(x))
    out[[paste0(v, "_mean")]] <- if (all(is.na(x))) NA_real_ else
      mean(x, na.rm = TRUE)
  }
  out
}))
cohort_display <- cohort_summary[, c("group", "n", "female")]
for (v in summary_vars) {
  cohort_display[[v]] <- sprintf("%.1f (n=%d)",
    cohort_summary[[paste0(v, "_mean")]], cohort_summary[[paste0(v, "_n")]])
}
knitr::kable(cohort_display, row.names = FALSE,
  caption = "Cohort summary: mean and available n for each measure; SI = insulin
  ↪  sensitivity.")
```

Table 1: Cohort summary: mean and available n for each measure; SI = insulin sensitivity.

group	n	female	age	bmi	hba1c	si
FH-	15	8	37.9 (n=15)	25.1 (n=15)	5.0 (n=15)	7.0 (n=15)
FH+	25	14	37.6 (n=25)	28.0 (n=25)	5.1 (n=25)	4.1 (n=25)
T2D	10	6	50.3 (n=10)	32.8 (n=10)	7.0 (n=10)	1.8 (n=8)

```
write.csv(cohort_summary, "results/cohort_summary.csv", row.names = FALSE)
kruskal.test(age ~ group, data = ph)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  age by group
## Kruskal-Wallis chi-squared = 7.595, df = 2, p-value = 0.02243
```

```
colSums(is.na(ph[, c("whr", "si", "glu2h")]))
```

```
##   whr    si glu2h
##     1     2    10
```

Data issues found. (i) The T2D group is older than the FH- and FH+ groups, so age and diabetes are associated in this sample and can confound expression comparisons (H4). (ii) OGTT values are missing for all T2D subjects, SI for two, and waist-hip ratio for one. The T2D SI mean uses its eight observed values; missing values do not erase the group mean. (iii) One FH- subject has BMI 20 but WHR 1.36, an unusual combination requiring source verification. It is not sufficient evidence of an entry error; WHR is not used for inference.

2.1 Quality control

```
ord <- order(ph$group)
boxplot(expr[, ord], outline = FALSE, las = 2, cex.axis = 0.5, names =
↪  ph$label[ord],
        col = c("grey85", "lightblue", "salmon")[ph$group[ord]], ylab = "log
        ↪  expression")
legend("topright", fill = c("grey85", "lightblue", "salmon"), legend =
↪  levels(ph$group),
       cex = 0.7, horiz = TRUE, bty = "n")
```

```
sexchk <- data.frame(XIST = expr["XIST", ], RPS4Y1 = expr["RPS4Y1", ], sex = ph$sex)
ggplot(sexchk, aes(XIST, RPS4Y1, colour = sex)) + geom_point(size = 2)
```

```
pred_sex <- ifelse(sexchk$XIST > 11 & sexchk$RPS4Y1 < 12, "female",
             ifelse(sexchk$XIST < 11 & sexchk$RPS4Y1 > 12, "male", "ambiguous"))
mm <- pred_sex != as.character(ph$sex)
knitr::kable(data.frame(sample = ph$label, reported = ph$sex, XIST = sexchk$XIST,
                        RPS4Y1 = sexchk$RPS4Y1, expression_says = pred_sex)[mm, ],
```

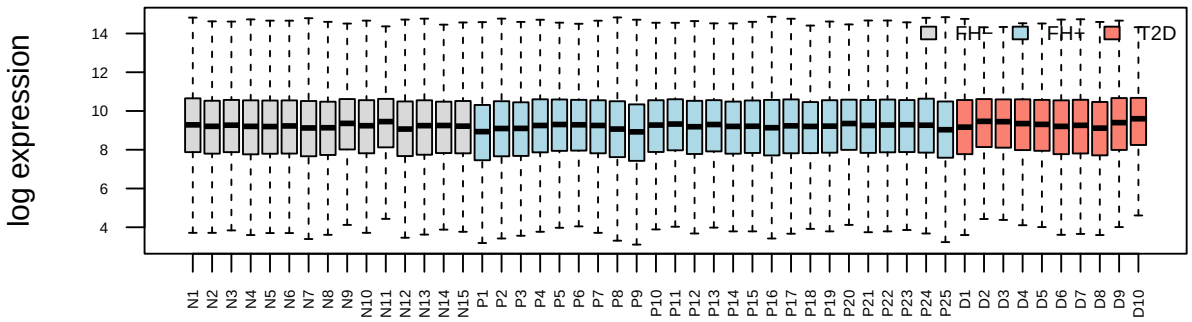


Figure 1: Per-sample distributions of the course-provided log-normalized matrix; no additional normalization was applied.

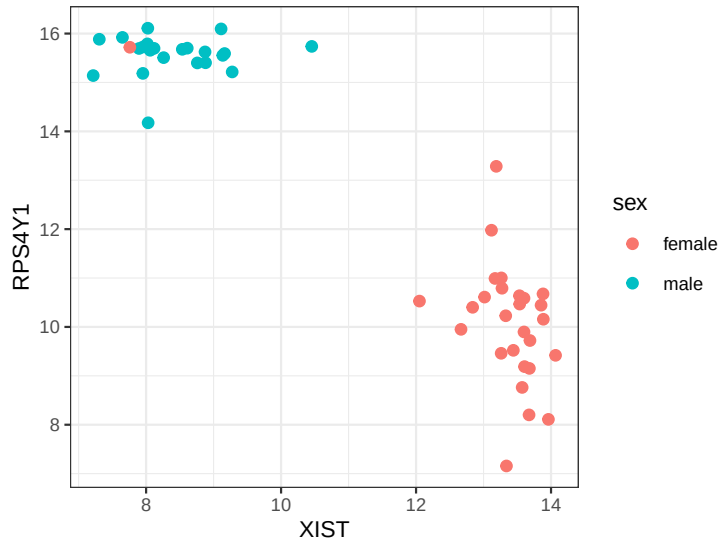


Figure 2: Sex check: XIST (X-inactivation) versus RPS4Y1 (Y-linked).

```
            digits = 1, row.names = FALSE, caption = "Samples whose expression
            ↪  disagrees with reported sex.")
```

Table 2: Samples whose expression disagrees with reported sex.

sample	reported	XIST	RPS4Y1	expression_says
N1	female	13.2	13.3	ambiguous
P25	female	7.8	15.7	male

The exploratory XIST/RPS4Y1 thresholds flag 2 of 50 samples: N1, P25. These are expression–metadata discrepancies, not diagnoses of sample mislabeling, contamination, or a karyotype. The two markers and heuristic cutoffs are insufficient to determine the cause. Reported labels and all samples are retained in the main analysis; flagged samples are excluded only in explicitly labeled age and PCA sensitivity analyses. Multiple sex-chromosome markers and source records would be needed before relabeling or permanently excluding a sample.

2.2 Outlier samples

```
cc <- cor(expr)
diag(cc) <- NA_real_  # exclude self-correlation explicitly
medcor <- apply(cc, 2, median, na.rm = TRUE)
thr <- quantile(medcor, 0.25) - 1.5 * IQR(medcor)
outl <- ph$label[medcor < thr]
o2 <- order(medcor)
barplot(medcor[o2], names.arg = ph$label[o2], las = 2, cex.names = 0.5,
        col = ifelse(medcor[o2] < thr, "firebrick", "grey70"), ylim = c(0.75, 0.9),
        ↪  xpd = FALSE,
        ylab = "median r with other samples")
abline(h = thr, lty = 2)
```

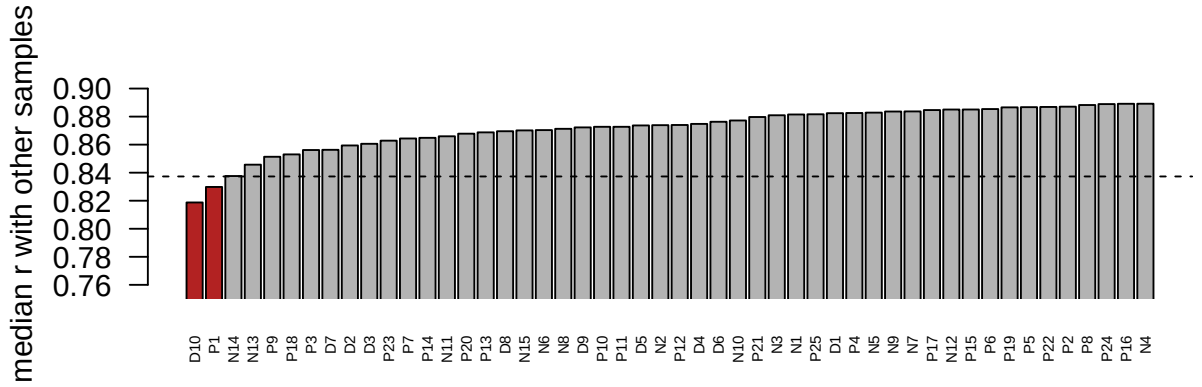


Figure 3: Median Pearson correlation of each sample with all other samples. Red: below Q1 - 1.5 IQR.

```
excl <- ph$sample[medcor < thr | mm]   # outliers + sex-discordant samples
```

The median between-sample correlation is 0.874. Lower-correlation flags: P1, D10. This within-dataset diagnostic does not establish technical failure or justify a numerical comparison with RNA-seq datasets processed on other scales. The union of correlation and sex-marker flags is N1, P1, D10, P25.

3 Heatmap of the muscle transcriptome

Genes were ranked by variance across samples and the 1,000 most variable were kept; each gene was z-scored across samples. Samples and genes were clustered hierarchically (Euclidean distance, Ward.D2 linkage).

```
v     <- apply(expr, 1, var)
top   <- names(sort(v, decreasing = TRUE))[1:1000]
z     <- t(scale(t(expr[top, ])))
ann   <- data.frame(Sex = ph$sex, Group = ph$group, Age = ph$age, BMI = ph$bmi,
                    row.names = ph$sample)
ann_col <- list(Sex = c(female = "#E7298A", male = "#1B9E77"),
                Group = c("FH-" = "grey70", "FH+" = "#7570B3", "T2D" = "#D95F02"))
```

```
pheatmap(z, show_rownames = FALSE, labels_col = ph$label, fontsize_col = 6,
         annotation_col = ann, annotation_colors = ann_col,
         clustering_method = "ward.D2", treeheight_row = 0,
         breaks = seq(-3, 3, length.out = 101),
         color = colorRampPalette(c("navy", "white", "firebrick3"))(100))
```

To see what lies under the sex signal, sex-associated genes were identified with a Welch t-test (BH q < 0.05 and |mean difference| > 1 log unit) and removed, and the heatmap was redrawn on the 1,000 most variable remaining genes.

```
tt <- apply(expr, 1, function(g) t.test(g ~ ph$sex)$p.value)
dm <- rowMeans(expr[, ph$sex == "male"]) - rowMeans(expr[, ph$sex == "female"])
sex_genes <- names(which(p.adjust(tt, "BH") < 0.05 & abs(dm) > 1))
length(sex_genes); head(sort(sex_genes), 25)
```

```
## [1] 19
```

```
##  [1] "ADH1B"        "CD24"         "CYTH2"        "DDX3Y"        "EIF1AY"
##  [6] "IRX3"         "KDM5D"        "LEP"          "LINC00844"    "LOC100506563"
## [11] "RPS4Y1"       "TPRG1"        "TSIX"         "TTTY14"       "TXLNGY"
## [16] "USP9Y"        "UTY"          "XIST"         "ZFY"
```

```
sex_variance_fraction <- sum(apply(expr[sex_genes, , drop = FALSE], 1, var)) /
  sum(apply(expr, 1, var))
expr_ns <- expr[setdiff(rownames(expr), sex_genes), ]
top_ns  <- names(sort(apply(expr_ns, 1, var), decreasing = TRUE))[1:1000]
z_ns    <- t(scale(t(expr_ns[top_ns, ])))
```

```
pheatmap(z_ns, show_rownames = FALSE, labels_col = ph$label, fontsize_col = 6,
         annotation_col = ann, annotation_colors = ann_col,
         clustering_method = "ward.D2", treeheight_row = 0,
         breaks = seq(-3, 3, length.out = 101),
```

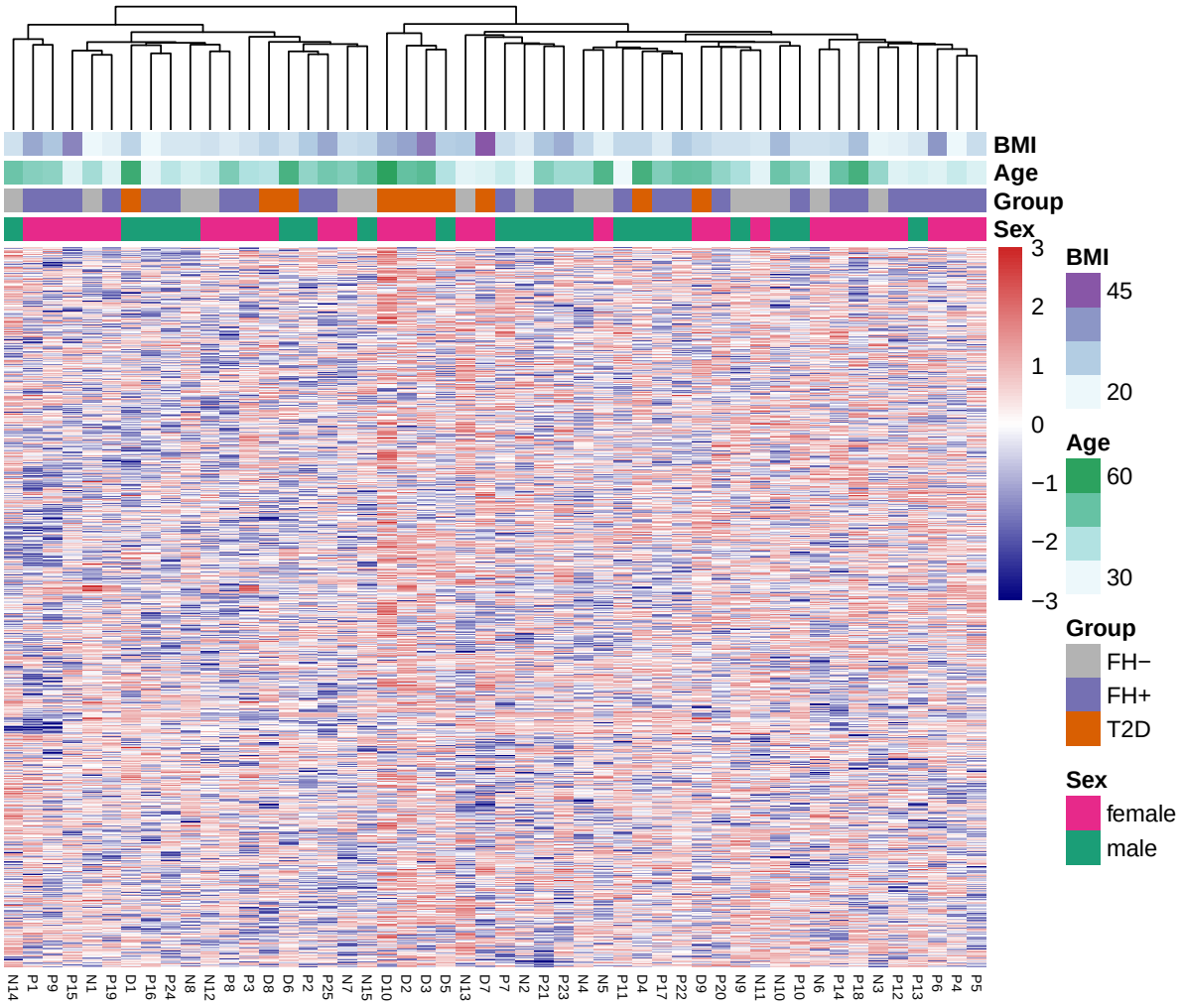


Figure 4: Top 1,000 most variable genes, z-scored, Ward.D2 clustering.

```
        color = colorRampPalette(c("navy", "white", "firebrick3"))(100))
```

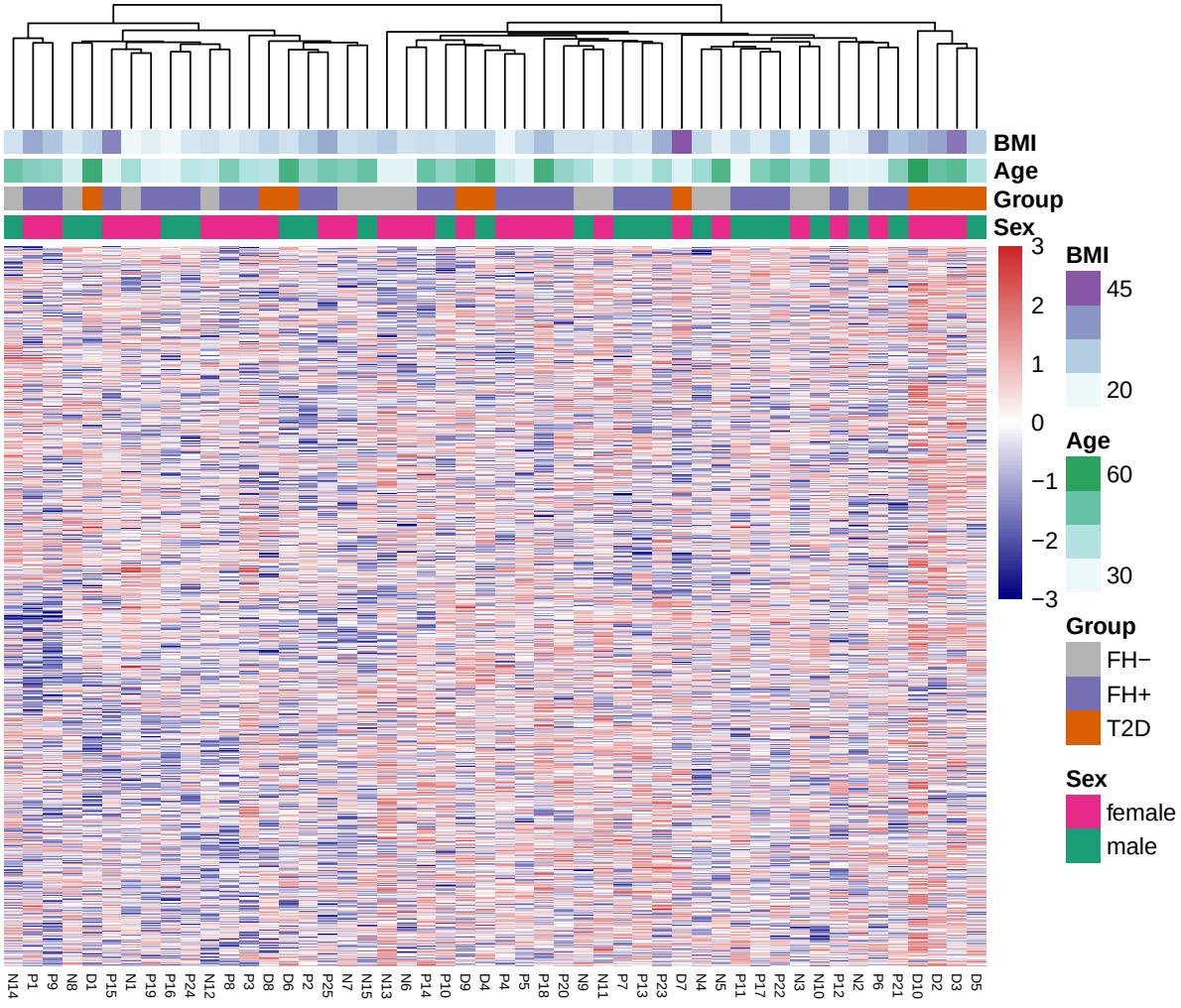


Figure 5: Top 1,000 most variable genes after removing sex-associated genes.

4 Clustering patients

Both k-means (25 random starts) and hierarchical clustering (Ward.D2) were applied to the samples, on the z-scored top-1,000 features with and without sex-associated features. The number of clusters was chosen by average silhouette width for k = 2 to 6; a final k-means fit at the selected k uses 25 random starts. Final-fit silhouette widths are reported separately from the candidate-k fits. Agreement between methods is measured with the adjusted Rand index (ARI). A best k among 2–6 does not itself establish discrete biological subgroups.

```
set.seed(700)  # isolate clustering from earlier plotting/interactive RNG state
ari <- function(a, b) {                          # adjusted Rand index
  tab <- table(a, b); n <- sum(tab); c2 <- function(x) sum(choose(x, 2))
  e <- c2(rowSums(tab)) * c2(colSums(tab)) / choose(n, 2)
  (c2(tab) - e) / ((c2(rowSums(tab)) + c2(colSums(tab))) / 2 - e)
}
```

```r
run_clusters <- function(zmat) {
  X  <- t(zmat); d <- dist(X); hc <- hclust(d, "ward.D2")
  sil <- sapply(2:6, function(k) {
    km <- kmeans(X, k, nstart = 25)
    c(kmeans = mean(silhouette(km$cluster, d)[, 3]),
      hclust = mean(silhouette(cutree(hc, k), d)[, 3]))
  })
  colnames(sil) <- paste0("k=", 2:6)
  list(X = X, d = d, hc = hc, sil = sil)
}
cl_all <- run_clusters(z); cl_ns <- run_clusters(z_ns)
round(cl_all$sil, 3); round(cl_ns$sil, 3)
```

```
##          k=2   k=3   k=4   k=5   k=6
## kmeans 0.021 0.019 0.018 0.014 0.015
## hclust 0.021 0.020 0.017 0.013 0.011
```

```
##          k=2   k=3   k=4   k=5   k=6
## kmeans 0.021 0.021 0.019 0.016 0.013
## hclust 0.021 0.020 0.017 0.012 0.013
```

```r
k_all <- which.max(cl_all$sil["kmeans", ]) + 1
k_ns  <- which.max(cl_ns$sil["kmeans", ]) + 1
ph$km_all <- factor(kmeans(cl_all$X, k_all, nstart = 25)$cluster)
ph$hc_all <- factor(cutree(cl_all$hc, k_all))
ph$km_ns  <- factor(kmeans(cl_ns$X, k_ns, nstart = 25)$cluster)
ph$hc_ns  <- factor(cutree(cl_ns$hc, k_ns))
c(k_all = k_all, ARI_all = round(ari(ph$km_all, ph$hc_all), 2),
  k_nosex = k_ns, ARI_nosex = round(ari(ph$km_ns, ph$hc_ns), 2))
```

```
##   k_all.k=2     ARI_all k_nosex.k=2   ARI_nosex
##        2.00        0.57        2.00        0.57
```

```r
table(kmeans_all = ph$km_all, sex = ph$sex)
```

```
##           sex
## kmeans_all female male
##          1     13   12
##          2     15   10
```

```r
table(kmeans_nosex = ph$km_ns, group = ph$group)
```

```
##             group
## kmeans_nosex FH- FH+ T2D
##            1   6  12   7
##            2   9  13   3
```

```r
final_silhouette <- c(
  all = mean(silhouette(as.integer(ph$km_all), cl_all$d)[, 3]),
  nosex = mean(silhouette(as.integer(ph$km_ns), cl_ns$d)[, 3]))
```

```
final_silhouette
```

```
##        all      nosex
## 0.02111696 0.02133676
```

4.1 Are clusters enriched for phenotypes?

Categorical phenotypes were tested with Fisher's exact test and continuous phenotypes with the Kruskal-Wallis test; p-values were BH-adjusted within each clustering.

```
enrich <- function(cl) {
  cat_v <- c(sex = "sex", group = "group", diabetes = "diabetes")
  con_v <- c(age = "age", BMI = "bmi", HbA1c = "hba1c", SI = "si",
             fasting_insulin = "finsulin", fasting_glucose = "fglu")
  p <- c(sapply(cat_v, function(v) fisher.test(table(cl, ph[[v]]))$p.value),
         sapply(con_v, function(v) kruskal.test(ph[[v]] ~ cl)$p.value))
  data.frame(phenotype = names(p), p = signif(p, 2), q_BH = signif(p.adjust(p,
   ↪  "BH"), 2))
}
en <- merge(setNames(enrich(ph$km_all), c("phenotype", "p_all", "q_all")),
            setNames(enrich(ph$km_ns),  c("phenotype", "p_nosex", "q_nosex")))
knitr::kable(en[order(en$q_all), ], row.names = FALSE,
             caption = "Phenotype enrichment of k-means clusters, all genes vs sex
              ↪  genes removed.")
```

Table 3: Phenotype enrichment of k-means clusters, all genes vs sex genes removed.

phenotype	p_all	q_all	p_nosex	q_nosex
BMI	0.073	0.22	0.073	0.22
fasting_insulin	0.067	0.22	0.067	0.22
SI	0.033	0.22	0.033	0.22
diabetes	0.290	0.65	0.290	0.65
group	0.380	0.68	0.380	0.68
age	0.850	0.85	0.850	0.85
fasting_glucose	0.830	0.85	0.830	0.85
HbA1c	0.740	0.85	0.740	0.85
sex	0.780	0.85	0.780	0.85

```
write.csv(en, "results/cluster_enrichment.csv", row.names = FALSE)
```

```
long <- do.call(rbind, lapply(c("age", "bmi", "hba1c", "si"), function(v)
  data.frame(cluster = ph$km_ns, variable = v, value = ph[[v]])))
ggplot(long, aes(cluster, value, fill = cluster)) + geom_boxplot(outlier.shape = NA)
 ↪  +
  geom_jitter(width = 0.15, size = 0.8) + facet_wrap(~variable, scales = "free_y",
   ↪  nrow = 1) +
  theme(legend.position = "none")
```

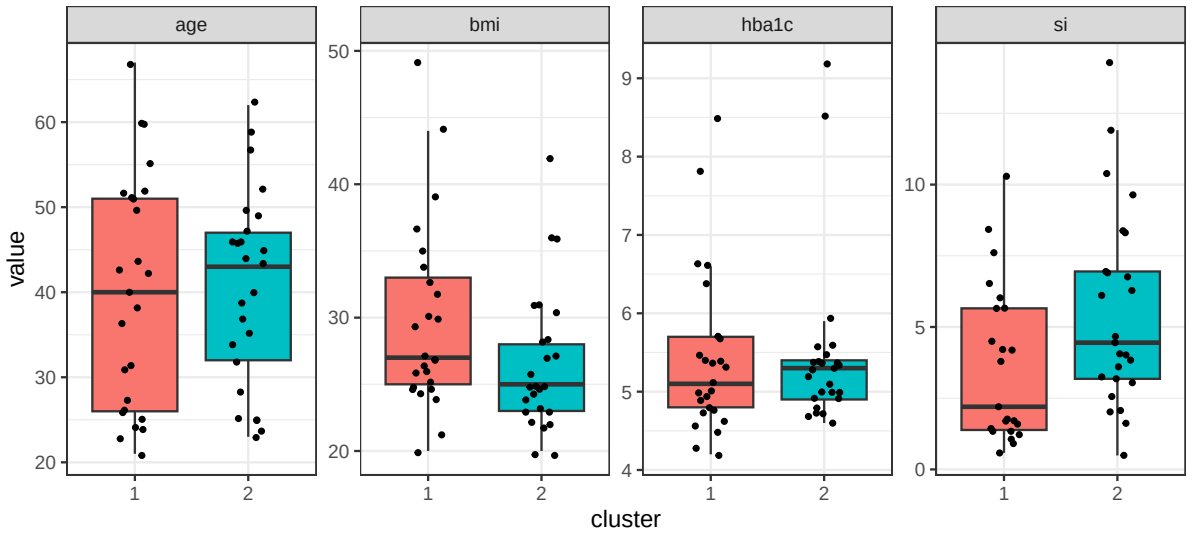


Figure 6: Phenotypes by k-means cluster (sex-associated genes removed).

Do male and female patients form distinct clusters? With all genes, the k-means partition (k = 2) against sex has Fisher p = 0.78. After sex-associated genes are removed, that association becomes p = 0.78.

5 Genes correlated with age

For every gene, the Pearson and Spearman correlation with age was computed across all 50 subjects, with Benjamini-Hochberg FDR. Because the T2D group is older (H4), two robustness checks were added: (a) a linear model `expression ~ age + sex + diabetes` fitted to every gene, and (b) the correlation re-computed in the 40 non-diabetic subjects only.

```
cor_test_vec <- function(M, x, method = "pearson") {
  use <- is.finite(x)
  M <- M[, use, drop = FALSE]; x <- x[use]
  stopifnot(ncol(M) > 2, all(is.finite(M)))
  r <- as.vector(cor(t(M), x, method = method))
  n <- length(x)
  tstat <- r * sqrt((n - 2) / (1 - r^2))
  p <- 2 * pt(-abs(tstat), n - 2)
  data.frame(gene = rownames(M), r = r, p = p, q = p.adjust(p, "BH"))
}
# Classical OLS fitted by QR; returns a specified coefficient for every feature.
fit_feature_lm <- function(M, metadata, formula, term) {
  X <- model.matrix(formula, metadata)
  stopifnot(nrow(X) == ncol(M), all(is.finite(X)), all(is.finite(M)))
  fit <- lm.fit(X, t(M))
  stopifnot(fit$rank == ncol(X), nrow(X) > fit$rank)
  j <- match(term, colnames(X)); stopifnot(!is.na(j))
  unscaled <- chol2inv(qr.R(fit$qr))
  j_pivot <- match(j, fit$qr$pivot)
```

```
  residual_df <- nrow(X) - fit$rank
  s2 <- colSums(fit$residuals^2) / residual_df
  se <- sqrt(s2 * unscaled[j_pivot, j_pivot])
  beta <- as.vector(fit$coefficients[j, ])
  p <- 2 * pt(-abs(beta / se), residual_df)
  data.frame(gene = rownames(M), beta = beta, se = se,
             p = p, q = p.adjust(p, "BH"), row.names = NULL)
}
pear <- cor_test_vec(expr, ph$age, "pearson")
spear <- cor_test_vec(expr, ph$age, "spearman")
nd <- ph$diabetes == "No"
pear_nd <- cor_test_vec(expr[, nd], ph$age[nd], "pearson")
age_adjusted <- fit_feature_lm(expr, ph, ~ age + sex + diabetes, "age")
age_tab <- data.frame(gene = pear$gene, r_pearson = pear$r,
  p_pearson = pear$p, q_pearson = pear$q,
  r_spearman = spear$r, p_spearman = spear$p, q_spearman = spear$q,
  r_nondiabetic = pear_nd$r, p_nondiabetic = pear_nd$p,
  beta_age_adj = age_adjusted$beta, p_adj = age_adjusted$p,
  q_adj = age_adjusted$q)
age_tab <- age_tab[order(age_tab$p_pearson), ]
write.csv(age_tab, "results/age_correlation_all_genes.csv", row.names = FALSE)
age_sig_counts <- c(
  pearson_q05 = sum(age_tab$q_pearson < 0.05),
  spearman_q05 = sum(age_tab$q_spearman < 0.05),
  adjusted_q05 = sum(age_tab$q_adj < 0.05),
  nondiabetic_p001 = sum(age_tab$p_nondiabetic < 0.001),
  pearson_p001 = sum(pear$p < 0.001))
age_sig_counts
```

```
##      pearson_q05     spearman_q05     adjusted_q05 nondiabetic_p001
##                0                1                1               35
##     pearson_p001
##               53
```

```
# Age-gene set: Pearson FDR < 0.05; fall back to the top 100 by p if fewer than 30
↪  pass
age_genes <- if (age_sig_counts["pearson_q05"] >= 30) age_tab$gene[age_tab$q_pearson
↪  < 0.05] else
  head(age_tab$gene, 100)
age_rule  <- if (age_sig_counts["pearson_q05"] >= 30) "Pearson FDR < 0.05" else "top
↪  100 by Pearson p"
robust    <- age_tab$gene[age_tab$gene %in% age_genes & age_tab$q_adj < 0.1 &
                          sign(age_tab$r_nondiabetic) == sign(age_tab$r_pearson)]
c(n_age_genes = length(age_genes), n_up = sum(age_tab$r_pearson[age_tab$gene %in%
↪  age_genes] > 0),
  n_robust = length(robust))
```

```
## n_age_genes        n_up    n_robust
##         100          44           1
```

```
knitr::kable(head(age_tab[, c("gene", "r_pearson", "q_pearson", "r_spearman",
                              "r_nondiabetic", "q_adj")], 20),
             digits = 3, row.names = FALSE,
             caption = "Top 20 age-correlated genes. q_adj: FDR for age in
             ↪  expression ~ age + sex + diabetes.")
```

Table 4: Top 20 age-correlated genes. q_adj: FDR for age in expression ~ age + sex + diabetes.

gene	r_pearson	q_pearson	r_spearman	r_nondiabetic	q_adj
UNC13C	0.612	0.051	0.705	0.638	0.043
MFF	-0.597	0.052	-0.591	-0.462	0.256
SCN4B	-0.578	0.083	-0.582	-0.464	0.266
SLIT2	0.544	0.188	0.549	0.417	0.419
PCDH9	0.541	0.188	0.517	0.469	0.203
C2orf88	-0.540	0.188	-0.513	-0.379	0.431
MRPL15	-0.534	0.203	-0.542	-0.347	0.401
DLAT	-0.525	0.210	-0.435	-0.342	0.466
RAVER2	-0.525	0.210	-0.516	-0.697	0.308
FAM96A	-0.523	0.210	-0.520	-0.407	0.438
AP2B1	-0.519	0.210	-0.534	-0.499	0.301
CADM2	-0.518	0.210	-0.516	-0.335	0.438
DCTPP1	-0.514	0.210	-0.479	-0.271	0.495
MDH2	-0.514	0.210	-0.537	-0.525	0.298
STRADB	-0.511	0.210	-0.493	-0.300	0.377
DSCAM-AS1	0.510	0.210	0.519	0.410	0.441
PTRHD1	-0.509	0.210	-0.544	-0.378	0.298
COMMD7	0.505	0.212	0.485	0.440	0.298
THNSL1	-0.504	0.212	-0.482	-0.464	0.431
ATP5F1	-0.504	0.212	-0.502	-0.403	0.438

The age-ranked set (top 100 by Pearson p) contains **100 expression features**. It is an exploratory selection, not a list of 100 FDR-significant genes. Full-sample counts at BH $q < 0.05$ are 0 (Pearson), 1 (Spearman), and 1 (age adjusted for sex and diabetes). Spearman p-values use the asymptotic t approximation, as in `cor.test(..., method = "spearman", exact = FALSE)`. Within the selected set, 1 feature meets the exploratory adjusted $q < 0.1$ rule and retains the Pearson direction in the overlapping non-diabetic subset. This is a within-cohort consistency check, not independent replication. The age-ordered heatmap and top-gene plots below visualize the selection and are not independent tests of it.

5.1 Sensitivity of age associations to flagged samples

```
keep_s <- !(ph$sample %in% excl)
age_sensitivity <- fit_feature_lm(
  expr[, keep_s], ph[keep_s, ], ~ age + sex + diabetes, "age")
age_sensitivity <- age_sensitivity[order(age_sensitivity$p), ]
write.csv(age_sensitivity, "results/age_adjusted_sample_sensitivity.csv",
```

```
          row.names = FALSE)
age_full_significant <- age_tab$gene[age_tab$q_adj < 0.05]
age_sens_significant <- age_sensitivity$gene[age_sensitivity$q < 0.05]
comparison_genes <- unique(c(age_full_significant, age_sens_significant))
age_sample_comparison <- data.frame(
  gene = comparison_genes,
  beta_all = age_tab$beta_age_adj[match(comparison_genes, age_tab$gene)],
  q_all = age_tab$q_adj[match(comparison_genes, age_tab$gene)],
  beta_without_flags = age_sensitivity$beta[
    match(comparison_genes, age_sensitivity$gene)],
  q_without_flags = age_sensitivity$q[
    match(comparison_genes, age_sensitivity$gene)])
knitr::kable(age_sample_comparison, digits = 4, row.names = FALSE,
  caption = "Same age + sex + diabetes model and feature universe: all 50 vs 46
  ↪  samples.")
```

Table 5: Same age + sex + diabetes model and feature universe: all 50 vs 46 samples.

gene	beta_all	q_all	beta_without_flags	q_without_flags
UNC13C	0.0408	0.0428	0.0414	0.0894
PLAG1	0.0647	0.2028	0.0773	0.0131

```
write.csv(age_sample_comparison, "results/age_sample_comparison.csv",
          row.names = FALSE)
```

After excluding N1, P1, D10, P25, the identical adjusted model and BH correction across all 22189 features identify **PLAG1** at $q < 0.05$ (0.0131). The full-sample significant feature is **UNC13C**. For UNC13C, adjusted q changes from 0.0428 to 0.0894. Thus the identity of the 5%-FDR age signal changes from UNC13C to PLAG1 when these samples are excluded. UNC13C still meets the exploratory 10% threshold after exclusion, but not 5%. This dependence on sample selection limits a claim of a stable age-gene signature; it does not prove that the flagged samples should be removed.

5.2 Heatmap of the exploratory age-ranked features

```
oa <- order(ph$age)
za <- t(scale(t(expr[age_genes, oa])))
pheatmap(za, cluster_cols = FALSE, show_rownames = length(age_genes) <= 60,
         fontsize_row = 5, labels_col = paste0(ph$label[oa], " (", ph$age[oa], ")"),
         fontsize_col = 5, annotation_col = ann, annotation_colors = ann_col,
         clustering_method = "ward.D2", treeheight_row = 20,
         breaks = seq(-3, 3, length.out = 101),
         color = colorRampPalette(c("navy", "white", "firebrick3"))(100))
```

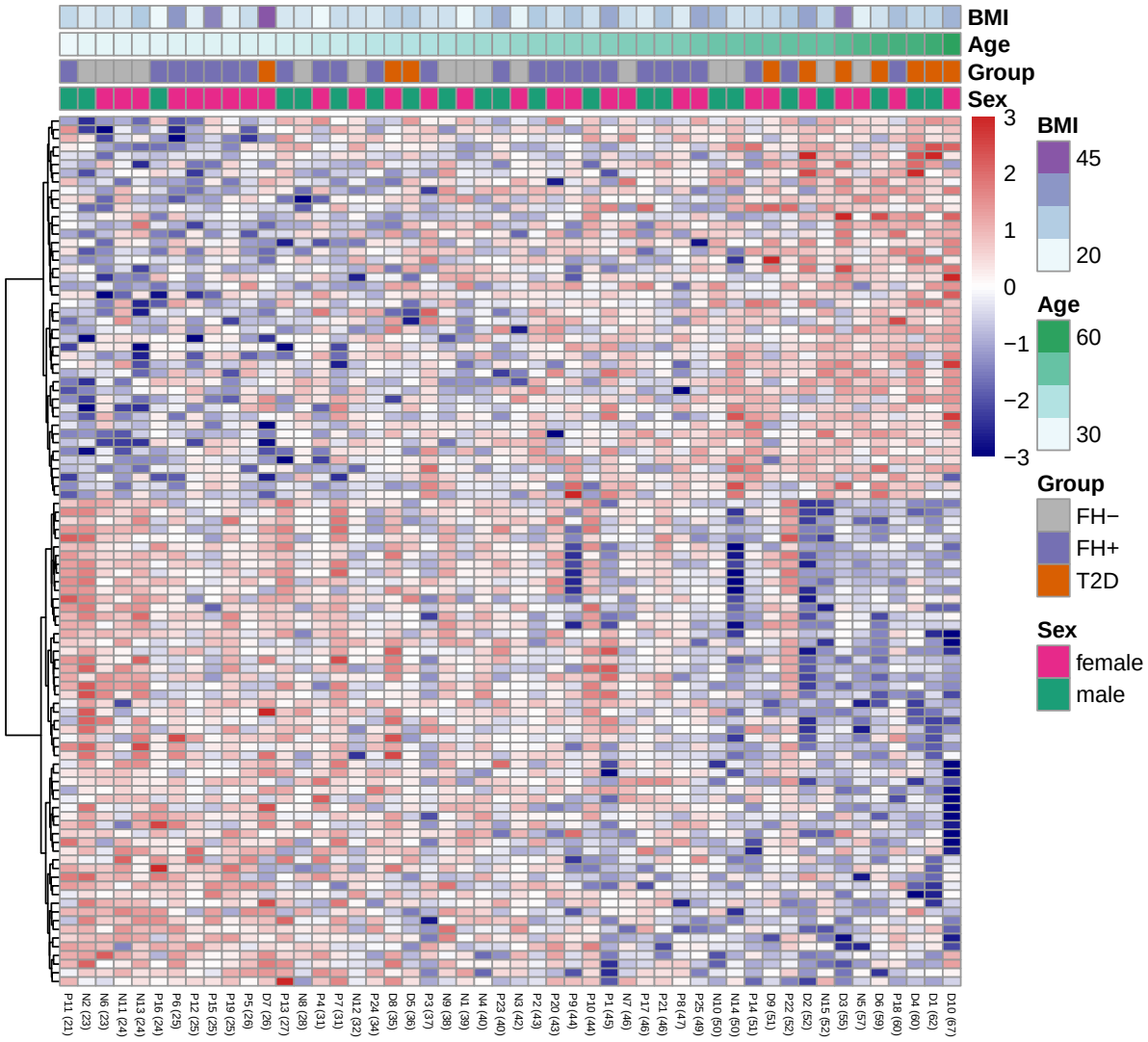


Figure 7: Exploratory age-ranked features; samples ordered by age (left = youngest). Rows clustered.

5.3 Correlation plots for the top age-correlated genes

```
top9 <- head(age_tab$gene, 9)
sc <- do.call(rbind, lapply(top9, function(g) data.frame(
  gene = sprintf("%s (r = %.2f)", g, age_tab$r_pearson[age_tab$gene == g]),
  age = ph$age, expr = expr[g, ], group = ph$group)))
sc$gene <- factor(sc$gene, levels = unique(sc$gene))
ggplot(sc, aes(age, expr)) + geom_point(aes(colour = group), size = 1.4) +
  geom_smooth(method = "lm", se = TRUE, colour = "black", linewidth = 0.6) +
  scale_colour_manual(values = ann_col$Group) +
  facet_wrap(~gene, scales = "free_y") + labs(x = "Age (years)", y = "log
  ↪  expression")
```

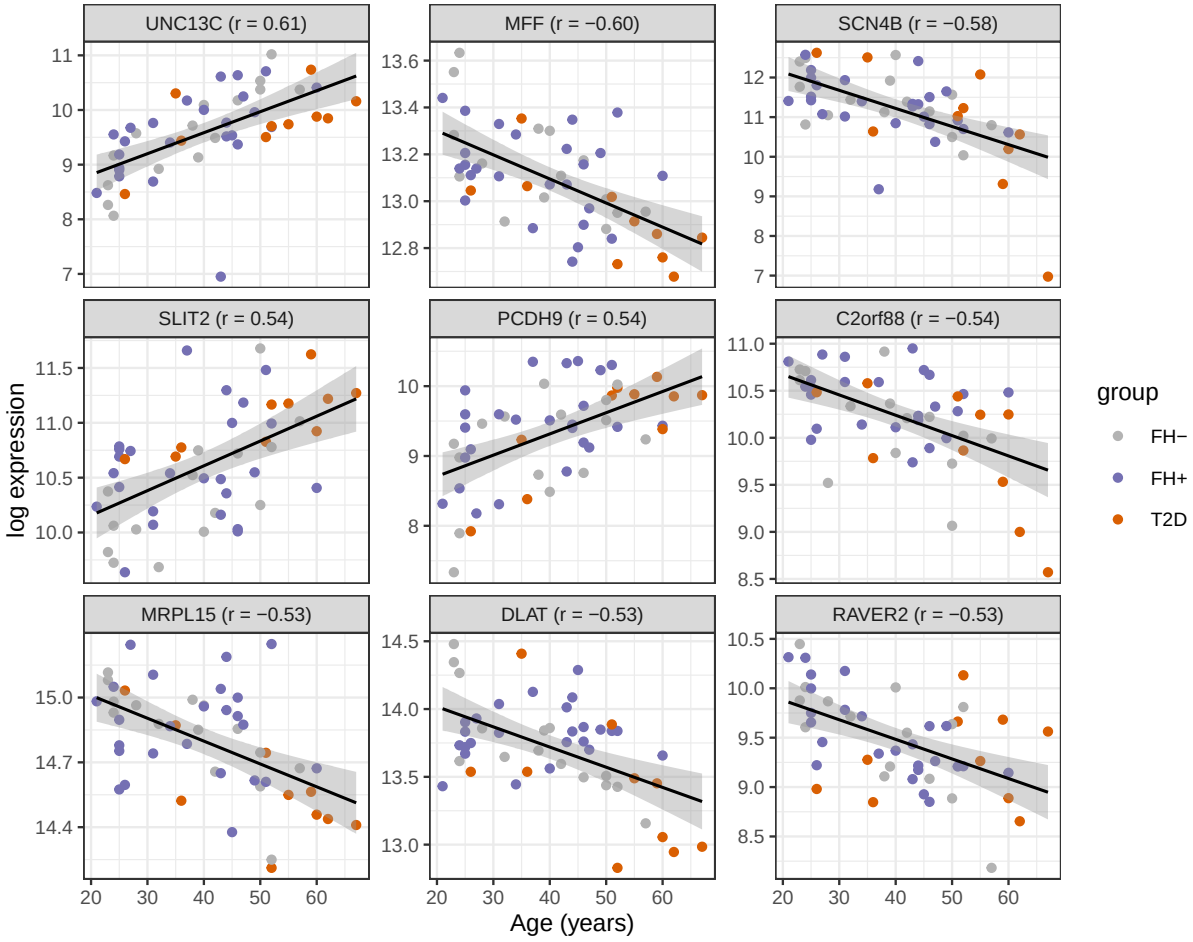


Figure 8: Top 9 age-correlated genes; T2D subjects in orange. Lines: linear fit on all subjects.

```
t30 <- head(age_tab, 30)
bar <- rbind(data.frame(gene = t30$gene, r = t30$r_pearson, set = "all (n = 50)"),
             data.frame(gene = t30$gene, r = t30$r_nondiabetic, set = "non-diabetic
             ↪  (n = 40)"))
bar$gene <- factor(bar$gene, levels = rev(t30$gene))
ggplot(bar, aes(r, gene, fill = set)) + geom_col(position = "dodge", width = 0.75) +
  geom_vline(xintercept = 0) + labs(x = "Pearson r with age", y = NULL) +
```

```
  theme(axis.text.y = element_text(size = 6), legend.position = "top")
```

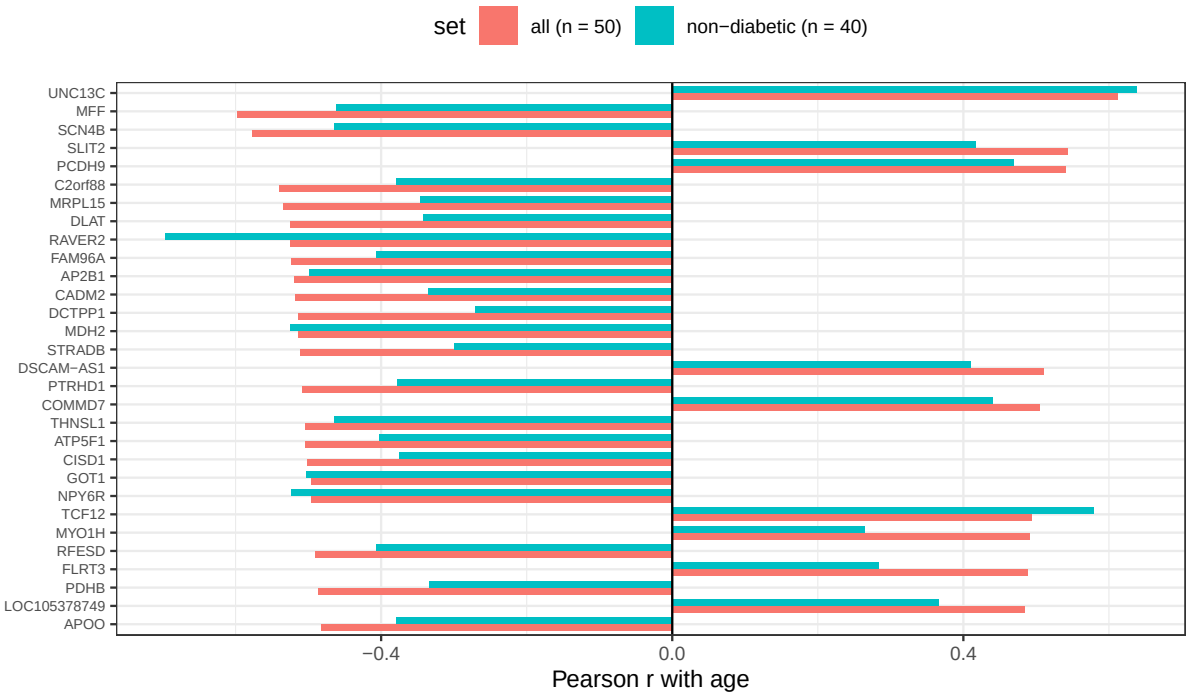


Figure 9: Top 30 age-correlated genes: Pearson r, all subjects vs non-diabetic subjects only.

6 Principal component analysis

PCA (`prcomp`) was run on all 22189 expression features, each centered to mean zero but not scaled (the data are already on a log scale, so unscaled PCA keeps high-variance genes influential). It was repeated after removing sex-associated genes. Each point is labeled with the subject code (N = FH-, P = FH+, D = T2D, followed by the GEO sample number within the group).

```
pca    <- prcomp(t(expr), center = TRUE, scale. = FALSE)
pca_ns <- prcomp(t(expr_ns), center = TRUE, scale. = FALSE)
ve     <- 100 * pca$sdev^2 / sum(pca$sdev^2)
ve_ns  <- 100 * pca_ns$sdev^2 / sum(pca_ns$sdev^2)
round(rbind(all = ve[1:6], no_sex = ve_ns[1:6]), 1)
```

```
##        [,1] [,2] [,3] [,4] [,5] [,6]
## all     7.6  4.4  3.9  3.5    3  2.8
## no_sex  7.6  4.4  3.9  3.5    3  2.8
```

```
pcd <- cbind(ph, pca$x[, 1:6]); pcd_ns <- cbind(ph, pca_ns$x[, 1:6])
```

```
ggplot(pcd, aes(PC1, PC2, colour = sex, shape = group)) + geom_point(size = 2.3) +
  geom_text_repel(aes(label = label), size = 2.4, max.overlaps = 50, show.legend =
  ↪  FALSE) +
  scale_colour_manual(values = ann_col$Sex) +
  labs(x = sprintf("PC1 (%.1f%%)", ve[1]), y = sprintf("PC2 (%.1f%%)", ve[2]))
```

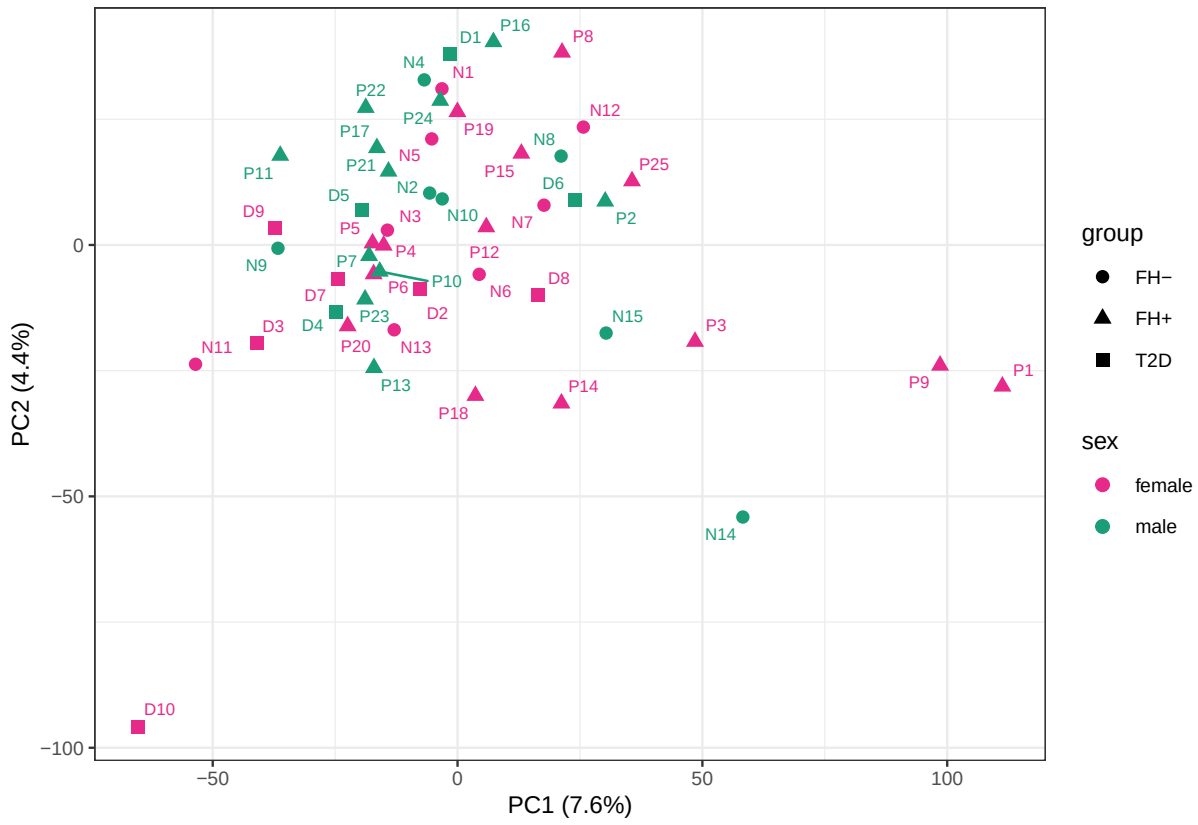


Figure 10: PCA of all genes, colored by sex, shaped by group.

```
ggplot(pcd_ns, aes(PC1, PC2, colour = age, shape = group)) + geom_point(size = 2.3)
↪  +
  geom_text_repel(aes(label = label), size = 2.4, max.overlaps = 50, show.legend =
  ↪  FALSE) +
  scale_colour_viridis_c() +
  labs(x = sprintf("PC1 (%.1f%%)", ve_ns[1]), y = sprintf("PC2 (%.1f%%)", ve_ns[2]))
```

Matched PCA sensitivity analysis. To isolate the effect of sample inclusion, the next two PCAs use the same 1,000 non-sex features selected in the full dataset and the same centering/unit-variance scaling rule. One includes all 50 samples, the other excludes the four exploratory flags. This pair separates sample inclusion from the different feature selection/scaling choices used by the preceding all-feature PCAs. PC axes are estimated anew in each fit; PC2 should not be assumed to represent an identical biological axis.

```
pca_matched <- prcomp(t(expr_ns[top_ns, ]), center = TRUE, scale. = TRUE)
ve_matched <- 100 * pca_matched$sdev^2 / sum(pca_matched$sdev^2)
pcd_matched <- cbind(ph, pca_matched$x[, 1:6])
ggplot(pcd_matched, aes(PC1, PC2, colour = group, size = bmi)) +
  geom_point(alpha = 0.85) +
  geom_text_repel(aes(label = label), size = 2.4, max.overlaps = 50,
                  show.legend = FALSE) +
  scale_colour_manual(values = ann_col$Group) + scale_size(range = c(1.2, 4)) +
  labs(x = sprintf("PC1 (%.1f%%)", ve_matched[1]),
```

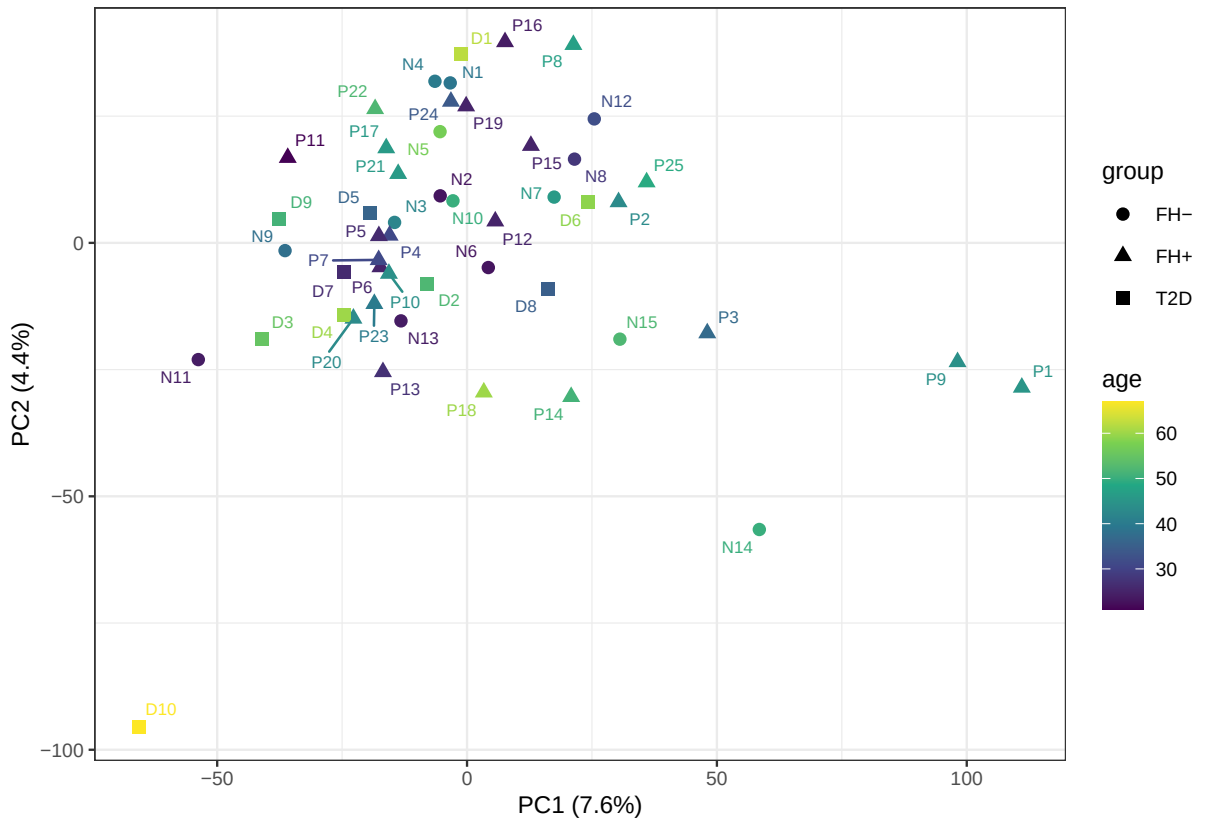


Figure 11: PCA after removing sex-associated genes, colored by age, shaped by group.

```
       y = sprintf("PC2 (%.1f%%)", ve_matched[2]))
```

```
keep_s  <- !(ph$sample %in% excl)
pca_s   <- prcomp(t(expr_ns[top_ns, keep_s]), center = TRUE, scale. = TRUE)
ve_s    <- 100 * pca_s$sdev^2 / sum(pca_s$sdev^2)
pcd_s   <- cbind(ph[keep_s, ], pca_s$x[, 1:6])
ggplot(pcd_s, aes(PC1, PC2, colour = group, size = bmi)) + geom_point(alpha = 0.85)
↪  +
  geom_text_repel(aes(label = label), size = 2.4, max.overlaps = 50, show.legend =
  ↪  FALSE) +
  scale_colour_manual(values = ann_col$Group) + scale_size(range = c(1.2, 4)) +
  labs(x = sprintf("PC1 (%.1f%%)", ve_s[1]), y = sprintf("PC2 (%.1f%%)", ve_s[2]))
```

```
pc_assoc <- function(d, analysis) {
  vars <- c("age", "bmi", "hba1c", "si", "finsulin")
  do.call(rbind, lapply(paste0("PC", 1:6), function(pc) {
    p <- c(sex = wilcox.test(d[[pc]] ~ d$sex)$p.value,
      group = kruskal.test(d[[pc]] ~ d$group)$p.value,
      sapply(vars, function(v)
        cor.test(d[[pc]], d[[v]], method = "spearman", exact = FALSE)$p.value))
    n <- c(sex = sum(complete.cases(d[, c(pc, "sex")])),
      group = sum(complete.cases(d[, c(pc, "group")])),
```

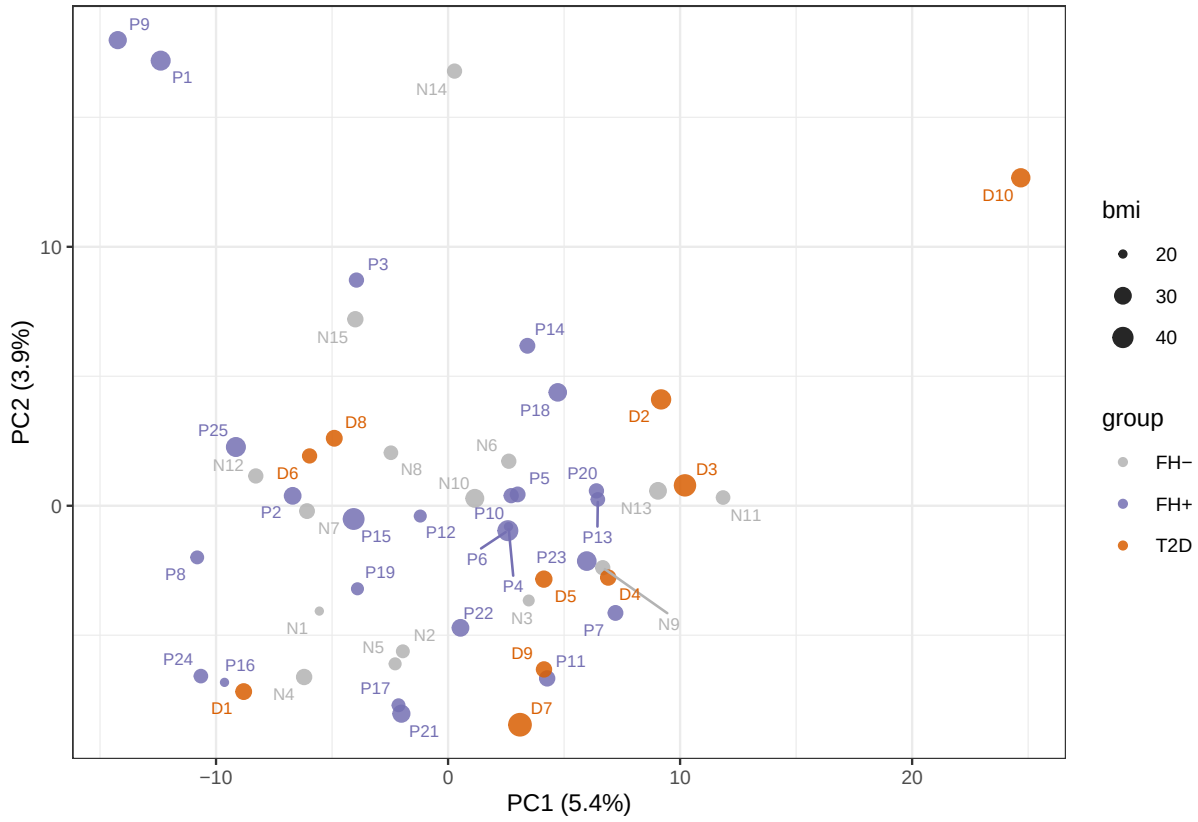


Figure 12: Matched PCA control: all 50 samples, fixed top 1,000 non-sex features, scaled. Compare with the next panel.

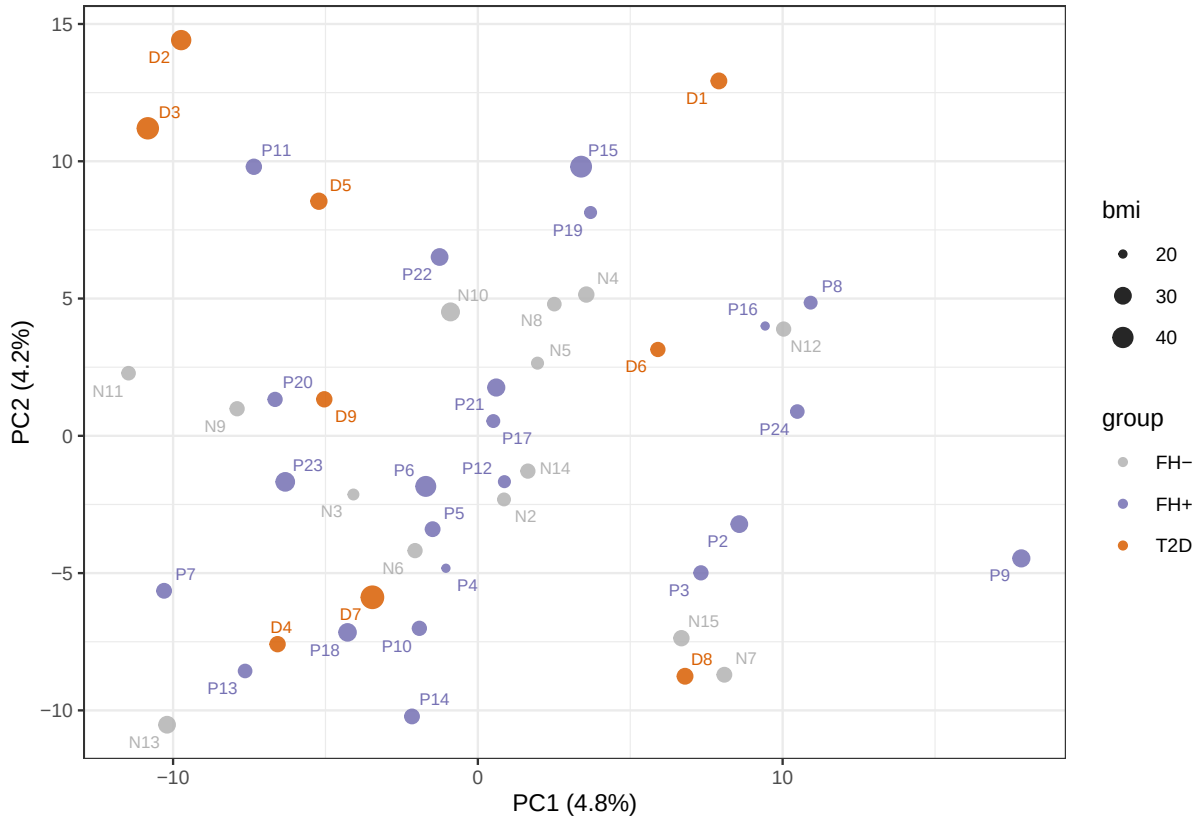


Figure 13: Matched PCA sensitivity: 46 samples after excluding exploratory QC flags; same top 1,000 non-sex features and scaling rule. Color: group; size: BMI.

```
      sapply(vars, function(v) sum(complete.cases(d[, c(pc, v)]))))
    data.frame(analysis = analysis, PC = pc, phenotype = names(p),
               n = as.integer(n), p = as.numeric(p))
  }))
}
pc_associations <- rbind(pc_assoc(pcd, "all_features"),
  pc_assoc(pcd_ns, "nonsex_features"),
  pc_assoc(pcd_matched, "matched_50"), pc_assoc(pcd_s, "matched_46"))
# Each of the four declared analyses has 6 PCs x 7 phenotypes = 42 tests.
pc_associations$q_within <- ave(pc_associations$p, pc_associations$analysis,
                                FUN = function(p) p.adjust(p, "BH"))
pc_associations$q_all <- p.adjust(pc_associations$p, "BH")
write.csv(pc_associations, "results/pc_phenotype_associations.csv",
          row.names = FALSE)
pc_nominal <- pc_associations[pc_associations$p < 0.05, ]
knitr::kable(pc_nominal, digits = 4, row.names = FALSE,
  caption = "Nominal p < 0.05 rows shown after BH correction of all 42 tests per PCA
  ↪  and all 168 tests together. Full results are exported.")
```

Table 6: Nominal p < 0.05 rows shown after BH correction of all 42 tests per PCA and all 168 tests together. Full results are exported.

analysis	PC	phenotype	n	p	q_within	q_all
all_features	PC2	bmi	50	0.0169	0.3648	0.4074
all_features	PC5	age	50	0.0488	0.3767	0.4614
all_features	PC5	hba1c	50	0.0319	0.3648	0.4461
all_features	PC6	sex	50	0.0299	0.3648	0.4461
all_features	PC6	finsulin	50	0.0347	0.3648	0.4490
nonsex_features	PC2	bmi	50	0.0188	0.3752	0.4074
nonsex_features	PC5	age	50	0.0415	0.3752	0.4614
nonsex_features	PC5	hba1c	50	0.0375	0.3752	0.4501
nonsex_features	PC6	group	50	0.0447	0.3752	0.4614
nonsex_features	PC6	finsulin	50	0.0277	0.3752	0.4461
matched_50	PC2	sex	50	0.0105	0.2780	0.4074
matched_50	PC4	sex	50	0.0132	0.2780	0.4074
matched_46	PC2	si	44	0.0178	0.2037	0.4074
matched_46	PC2	finsulin	46	0.0306	0.2574	0.4461
matched_46	PC3	sex	46	0.0150	0.2037	0.4074
matched_46	PC6	group	46	0.0194	0.2037	0.4074
matched_46	PC6	si	44	0.0187	0.2037	0.4074

```
pc2_sensitivity <- pc_associations[
  pc_associations$analysis == "matched_46" & pc_associations$PC == "PC2", ]
```

There are 168 PC–phenotype tests in total. Within-PCA BH q < 0.05 is reached by 0 tests; across-all-analyses BH q < 0.05 by 0. In the 46-sample PCA, the PC2–SI association has p = 0.0178 and

within-PCA q = 0.204. It is a nominal exploratory association, not an FDR-significant finding.

```
ld <- pca_ns$rotation
data.frame(PC1_top_pos = names(head(sort(ld[, 1], decreasing = TRUE), 10)),
           PC1_top_neg = names(head(sort(ld[, 1]), 10)),
           PC2_top_pos = names(head(sort(ld[, 2], decreasing = TRUE), 10)),
           PC2_top_neg = names(head(sort(ld[, 2]), 10)))
```

```
##        PC1_top_pos       PC1_top_neg             PC2_top_pos  PC2_top_neg
## 1            RPS11              PFN1                   CRIP2 LOC101928865
## 2           ADIPOQ             HSPB6                     DDN      C7orf57
## 3           FRG1HP            COX6A2                   ASCC2        DDX53
## 4            PLAG1              SELM                   BRAT1 LOC100996325
## 5  CCT6P1///CCT6P3            IFITM3                   UBE2M        RPS11
## 6            PGBD1       HBA2///HBA1                   HCP5      SLCO1B3
## 7          SLC2A10             HSPB1                 TMEM158     ANKRD30B
## 8              GZMM LAMA5-AS1///LAMA5       SNORA5B///TBRG4        TNIP3
## 9         LOC284926         LINC00116                 PPFIA4       B3GNT7
## 10          CCDC110             UBE2M LOC101060835///HLA-DQB1     CYP4F62P
```

7 Additional analysis: insulin sensitivity and diabetes

As an exploratory extension, every gene was correlated (Spearman) with insulin sensitivity (SI, n = 48), and T2D was compared with FH- subjects (Welch t-test). Genes from Jin et al.'s SRF story (SRF, its co-activator MKL1/MRTF-A, and the SRF targets ACTA2, FHL1, VCL, CNN2) are checked individually. These targeted checks reuse the original study's public cohort; they are not independent replication. The Welch contrast is unadjusted; an additional model below estimates the T2D–FH- difference conditional on age and reported sex.

```
si_ok <- !is.na(ph$si)
si_cor <- cor_test_vec(expr[, si_ok], ph$si[si_ok], "spearman")
si_cor <- si_cor[order(si_cor$p), ]
c(si_q05 = sum(si_cor$q < 0.05), si_p001 = sum(si_cor$p < 0.001))
```

```
##  si_q05 si_p001
##       0      50
```

```
write.csv(si_cor, "results/si_correlation_all_genes.csv", row.names = FALSE)

d_vs_n <- ph$group %in% c("T2D", "FH-")
grp <- droplevels(ph$group[d_vs_n])
de_p <- apply(expr[, d_vs_n], 1, function(g) t.test(g ~ grp)$p.value)
de_fc <- rowMeans(expr[, ph$group == "T2D"]) - rowMeans(expr[, ph$group == "FH-"])
de <- data.frame(gene = names(de_p), log_diff_T2D_vs_FHneg = de_fc, p = de_p,
                 q = p.adjust(de_p, "BH"))[order(de_p), ]
c(de_q05 = sum(de$q < 0.05), de_p001 = sum(de$p < 0.001))
```

```
##  de_q05 de_p001
##       0      44
```

```r
write.csv(de, "results/T2D_vs_FHneg.csv", row.names = FALSE)
de_adjusted <- fit_feature_lm(expr, ph, ~ age + sex + group, "groupT2D")
de_adjusted <- de_adjusted[order(de_adjusted$p), ]
write.csv(de_adjusted, "results/T2D_vs_FHneg_age_sex_adjusted.csv",
          row.names = FALSE)

srf <- intersect(c("SRF", "MKL1", "MRTFA", "ACTA2", "FHL1", "VCL", "CNN2", "MYH11"),
 ↪  rownames(expr))
knitr::kable(data.frame(gene = srf,
  r_SI = si_cor$r[match(srf, si_cor$gene)], p_SI = si_cor$p[match(srf,
   ↪  si_cor$gene)],
  diff_T2D_FHneg = de$log_diff_T2D_vs_FHneg[match(srf, de$gene)],
  p_T2D_FHneg = de$p[match(srf, de$gene)]), digits = 3, row.names = FALSE,
  caption = "Selected SRF-related genes: unadjusted associations and nominal
   ↪  p-values; these are not measurements of SRF transcriptional activity.")
```

Table 7: Selected SRF-related genes: unadjusted associations and nominal p-values; these are not measurements of SRF transcriptional activity.

gene	r_SI	p_SI	diff_T2D_FHneg	p_T2D_FHneg
SRF	-0.370	0.010	0.294	0.017
MKL1	-0.440	0.002	0.483	0.143
ACTA2	-0.233	0.111	0.195	0.199
FHL1	-0.234	0.109	0.420	0.070
VCL	-0.070	0.638	0.585	0.041
CNN2	-0.188	0.202	0.523	0.038
MYH11	-0.053	0.720	0.095	0.802

```r
c(overlap_age_SI = length(intersect(age_genes, head(si_cor$gene,
 ↪  length(age_genes)))))
```

```
## overlap_age_SI
##              1
```

```r
de$lab <- ifelse(rank(de$p) <= 15, de$gene, "")
ggplot(de, aes(log_diff_T2D_vs_FHneg, -log10(p))) +
  geom_point(size = 0.4, alpha = 0.4, colour = "grey40") +
  geom_point(data = de[de$lab != "", ], colour = "#D95F02", size = 1.2) +
  geom_text_repel(aes(label = lab), size = 2.5, max.overlaps = 30) +
  labs(x = "mean log expression, T2D minus FH-", y = "-log10 p")
```

The unadjusted Welch comparison yields 0 features at BH q < 0.05; the age/sex-adjusted T2D contrast yields 2. The latter includes all three groups in `expression ~ age + sex + group`, with FH- as reference. BMI is not included in this primary descriptive contrast: whether to adjust for adiposity depends on the intended estimand. Neither contrast identifies a causal diabetes effect, and residual confounding remains possible.

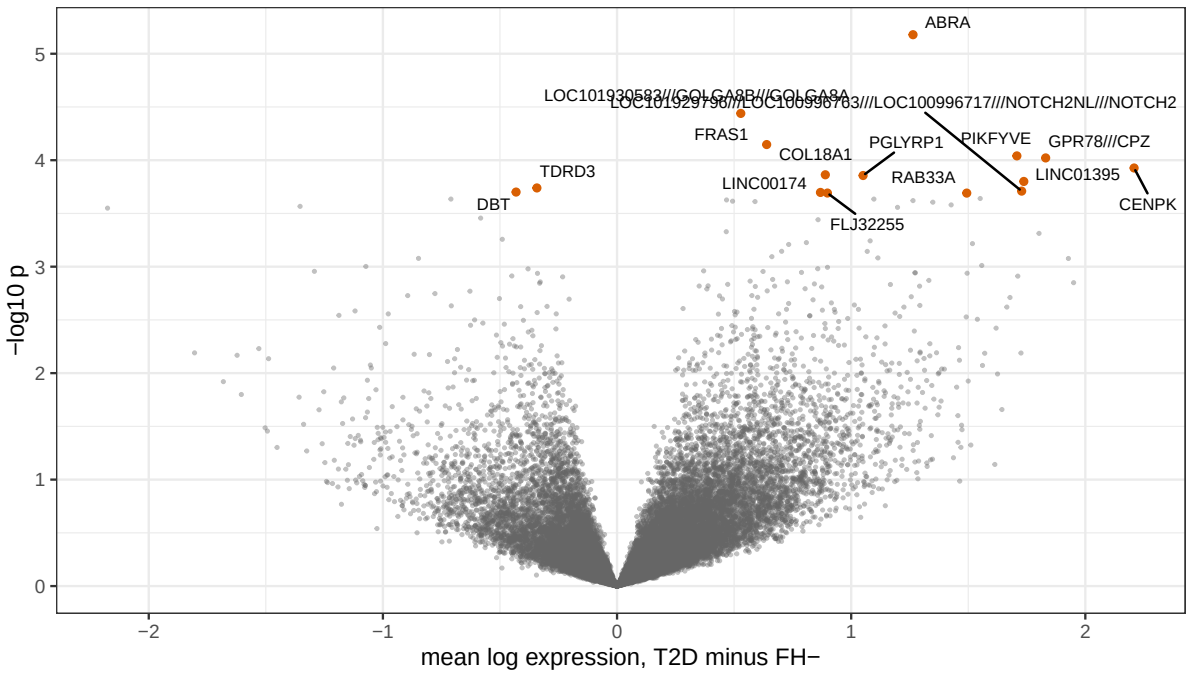


Figure 14: T2D vs FH-: difference in mean log expression vs significance.

8 DAVID functional enrichment (Part III)

Five gene lists are written to `results/` and analyzed with the DAVID Functional Annotation Chart (https://davidbioinformatics.nih.gov, run on October 21, 2026; the site listed Knowledgebase v2026_2 as its latest release): genes decreasing with age (`age_down`), genes increasing with age (`age_up`), the top 100 marker genes of each k-means cluster after sex-gene removal (`cluster1_up`, `cluster2_up`), and the 200 genes most correlated with insulin sensitivity (`SI_top200`). The single robust age gene was not submitted. Settings: identifier OFFICIAL_GENE_SYMBOL, species *Homo sapiens*, background = DAVID's built-in *Human Genome U133 Plus 2 Array* population (the array used here, so the background is the genes measurable on this platform), categories KEGG_PATHWAY and GOTERM_BP_DIRECT, default chart thresholds (Count >= 2, EASE p <= 0.1). Probe sets annotated with several symbols (`A///B`) were split into separate symbols before upload. DAVID is a web tool, so its output is stored unchanged in `data/david/` and read back here. DAVID's `Benjamini` column is the ordinary BH adjustment; its `FDR` column is the adaptive linear step-up adjustment described by Benjamini and Hochberg (2000). The reported enrichment threshold uses the latter. The platform background is appropriate to the supplied metadata; a separately justified analysis could instead use the actual tested/mappable universe. Multi-symbol features remain ambiguous measurements even after splitting symbols for annotation.

Cache validity: this document does not query DAVID automatically. Frozen normalized input lists are stored with the imported tables. A render must stop if the newly generated lists differ; changing a threshold, clustering result, or gene selection then requires a new DAVID run and a coordinated update of the frozen lists and results.

```
up   <- age_tab$gene[age_tab$gene %in% age_genes & age_tab$r_pearson > 0]
down <- age_tab$gene[age_tab$gene %in% age_genes & age_tab$r_pearson < 0]
writeLines(up,   "results/DAVID_age_up.txt")
writeLines(down, "results/DAVID_age_down.txt")
```

```
writeLines(robust, "results/DAVID_age_robust.txt")
writeLines(head(si_cor$gene, 200), "results/DAVID_SI_top200.txt")
writeLines(rownames(expr), "results/DAVID_background.txt")
# cluster marker genes: top 100 genes by t-statistic for each no-sex k-means
↪  cluster
for (k in levels(ph$km_ns)) {
  tk <- apply(expr_ns, 1, function(g) t.test(g[ph$km_ns == k], g[ph$km_ns !=
↪  k])$statistic)
  writeLines(names(head(sort(tk, decreasing = TRUE), 100)),
             sprintf("results/DAVID_cluster%s_up.txt", k))
}
c(age_up = length(up), age_down = length(down), robust = length(robust))
```

```
##   age_up age_down   robust
##       44       56        1
```

```
normalize_david_symbols <- function(x) {
  y <- toupper(trimws(unlist(strsplit(x, "///", fixed = TRUE))))
  unique(y[nzchar(y)])
}
check_david_inputs <- function(current, frozen) {
  if (!setequal(names(current), names(frozen)))
    stop("DAVID list names changed; rerun DAVID and update its frozen
    ↪  inputs/results.")
  changed <- names(current)[!vapply(names(current), function(n)
    setequal(current[[n]], frozen[[n]]), logical(1))]
  if (length(changed)) stop("Stale DAVID results for: ", paste(changed, collapse =
  ↪  ", "),
    ". Rerun DAVID and update its frozen inputs/results before rendering.")
  invisible(TRUE)
}
david_list_names <- c("age_down", "age_up",
  paste0("cluster", levels(ph$km_ns), "_up"), "SI_top200")
david_current_symbols <- setNames(lapply(david_list_names, function(n)
  normalize_david_symbols(readLines(paste0("results/DAVID_", n, ".txt")))),
  david_list_names)
david_frozen_names <- read.delim("data/david/DAVID_run_info.tsv")$List
david_frozen_symbols <- setNames(lapply(david_frozen_names, function(n)
  readLines(paste0("data/david/input_symbols/", n, ".txt"))), david_frozen_names)
check_david_inputs(david_current_symbols, david_frozen_symbols)
dv <- read.delim("data/david/DAVID_chart_results.tsv", check.names = FALSE)
dinfo <- read.delim("data/david/DAVID_run_info.tsv")
lists <- dinfo$List
dv$Category <- ifelse(dv$Category == "KEGG_PATHWAY", "KEGG", "GO-BP")
dv$Term <- sub("^(hsa[0-9]+:|GO:[0-9]+~)", "", dv$Term)
dinfo$terms_p_le_0.1 <- as.vector(table(factor(dv$List, lists)))
dinfo$terms_FDR_lt_0.05 <- as.vector(table(factor(dv$List[dv$FDR < 0.05], lists)))
knitr::kable(dinfo[, c("List", "input_symbols", "DAVID_IDs", "terms_p_le_0.1",
↪  "terms_FDR_lt_0.05")],
```

```
            row.names = FALSE,
            caption = "DAVID input lists: symbols submitted, symbols mapped to
            ↪  DAVID IDs, and KEGG + GO-BP terms returned.")
```

Table 8: DAVID input lists: symbols submitted, symbols mapped to DAVID IDs, and KEGG + GO-BP terms returned.

List	input_symbols	DAVID_IDs	terms_p_le_0.1	terms_FDR_lt_0.05
age_down	58	47	22	8
age_up	47	42	6	0
cluster1_up	106	92	18	0
cluster2_up	109	102	19	0
SI_top200	246	211	18	0

```
top <- do.call(rbind, lapply(split(dv, list(dv$List, dv$Category), drop = TRUE),
                             function(d) head(d[order(d$PValue), ], 3)))
top <- top[order(match(top$List, lists), top$Category != "KEGG", top$PValue), ]
top$Term <- ifelse(nchar(top$Term) > 48, paste0(substr(top$Term, 1, 45), "..."),
↪  top$Term)
top$fold <- round(top$`Fold Enrichment`, 1)
top$p <- signif(top$PValue, 2); top$FDR <- signif(top$FDR, 2)
knitr::kable(top[, c("List", "Category", "Term", "Count", "fold", "p", "FDR")],
↪  row.names = FALSE,
             caption = "Top 3 KEGG and top 3 GO-BP terms per list (DAVID EASE p; FDR
             ↪  < 0.05 only in age_down).")
```

Table 9: Top 3 KEGG and top 3 GO-BP terms per list (DAVID EASE p; FDR < 0.05 only in age_down).

List	Category	Term	Count	fold	p	FDR
age_down	KEGG	Carbon metabolism	5	15.9	2.0e-04	0.012
age_down	KEGG	Lipoic acid metabolism	3	59.8	1.0e-03	0.030
age_down	KEGG	Citrate cycle (TCA cycle)	3	35.9	2.8e-03	0.041
age_down	GO-BP	tricarboxylic acid cycle	4	50.2	6.2e-05	0.012
age_down	GO-BP	malate-aspartate shuttle	3	142.3	1.8e-04	0.018
age_down	GO-BP	mitochondrial fission	3	85.4	5.2e-04	0.034
age_up	GO-BP	mRNA splicing, via spliceosome	4	10.0	6.7e-03	0.960
age_up	GO-BP	RNA splicing	4	9.1	8.6e-03	0.960
age_up	GO-BP	regulation of neurogenesis	2	27.4	6.8e-02	1.000
cluster1_up	KEGG	EGFR tyrosine kinase inhibitor resistance	3	10.1	3.3e-02	1.000
cluster1_up	KEGG	Pathways in cancer	6	3.0	4.0e-02	1.000
cluster1_up	KEGG	Ras signaling pathway	4	4.6	5.1e-02	1.000
cluster1_up	GO-BP	lung development	4	25.7	4.8e-04	0.290
cluster1_up	GO-BP	embryonic organ development	3	24.1	6.6e-03	1.000
cluster1_up	GO-BP	in utero embryonic development	3	18.3	1.1e-02	1.000

List	Category	Term	Count	fold	p	FDR
cluster2_up	KEGG	Polycomb repressive complex	5	10.7	1.1e-03	0.092
cluster2_up	KEGG	Endocytosis	6	4.3	1.2e-02	0.500
cluster2_up	KEGG	Spliceosome	4	5.0	4.3e-02	1.000
cluster2_up	GO-BP	mRNA splicing, via spliceosome	6	5.2	5.7e-03	1.000
cluster2_up	GO-BP	multivesicular body assembly	3	16.5	1.4e-02	1.000
cluster2_up	GO-BP	lung development	3	15.1	1.6e-02	1.000
SI_top200	KEGG	Breast cancer	5	4.0	3.5e-02	1.000
SI_top200	KEGG	Endocrine resistance	4	4.8	5.0e-02	1.000
SI_top200	GO-BP	positive regulation of cell cycle process	3	33.3	3.4e-03	1.000
SI_top200	GO-BP	determination of left/right symmetry	4	12.0	4.4e-03	1.000
SI_top200	GO-BP	pH reduction	2	74.1	2.7e-02	1.000

```
david_sig_terms <- dv[dv$FDR < 0.05,
  c("List", "Category", "Term", "Count", "Genes", "FDR")]
david_sig_terms$FDR <- signif(david_sig_terms$FDR, 2)
knitr::kable(david_sig_terms[, setdiff(names(david_sig_terms), "List")],
             row.names = FALSE,
             caption = "All DAVID terms with FDR < 0.05 (age_down list) and the
             ↪  genes that drive them.")
```

Table 10: All DAVID terms with FDR < 0.05 (age_down list) and the genes that drive them.

Category	Term	Count	Genes	FDR
KEGG	Carbon metabolism	5	GCSH, GOT1, MDH2, DLAT, PDHB	0.012
KEGG	Lipoic acid metabolism	3	GCSH, DLAT, PDHB	0.030
KEGG	Citrate cycle (TCA cycle)	3	MDH2, DLAT, PDHB	0.041
KEGG	Metabolic pathways	11	GCSH, ALAS1, GOT1, MDH2, CA4, CISD1, L2HGDH, DLAT, PDHB, NDUFV3, DCTPP1	0.041
KEGG	2-Oxocarboxylic acid metabolism	3	GOT1, DLAT, PDHB	0.041
GO-BP	tricarboxylic acid cycle	4	MDH2, L2HGDH, DLAT, PDHB	0.012
GO-BP	malate-aspartate shuttle	3	GOT1, MDH2, SLC25A12	0.018
GO-BP	mitochondrial fission	3	MUL1, MTFR1L, MFF	0.034

Both the ordinary BH and adaptive-FDR columns identify 8 terms below 0.05 in this saved result; all are in the age-down list. This agreement does not make the two correction methods identical.

Interpretation. Only the age-decreasing list gives terms that survive adaptive FDR < 0.05, and several overlapping terms describe mitochondrial energy metabolism. KEGG carbon metabolism, lipoic acid metabolism and the citrate (TCA) cycle, and GO-BP tricarboxylic acid cycle, malate-aspartate shuttle and mitochondrial fission are all enriched, driven by a small core of genes (DLAT and PDHB of

pyruvate dehydrogenase, MDH2, GOT1, GCSH, L2HGDH, SLC25A12, and the fission genes MFF, MUL1 and MTFR1L). This is an exploratory functional theme relevant to the mitochondrial hallmark of aging; expression-based over-representation does not demonstrate mitochondrial dysfunction or an age-specific mechanism. Extracellular-matrix and inflammation terms are not significant in these selected lists, which does not establish absence of those processes. The genes increasing with age show only a nominal mRNA-splicing signal (FDR about 0.96). The two cluster marker lists give only nominal terms (cluster 1: growth-factor receptor signaling through PDGFRB, FGFR2 and VEGFA; cluster 2: Polycomb repressive complex, splicing and endocytosis, best FDR = 0.09). These are exploratory descriptions of expression-defined clusters, not independent validation of the clusters. The SI list has no pathway-level signal, so the SRF result above rests on the targeted gene tests, not on DAVID.

9 Discussion

Unsupervised structure and H1. Under the chosen feature selection, scaling, distance, and clustering methods, separation is weak: final silhouette widths are 0.021 and 0.021, and k-means/hierarchical ARI is 0.57. Selecting the best k among 2–6 does not prove that discrete biological subgroups exist. Cluster enrichment for SI is nominal ($p = 0.033$, $q = 0.22$); no tested phenotype survives within-clustering BH $q < 0.05$. H1 is not clearly supported. The SI/BMI/insulin trends are hypotheses for follow-up, not established drivers of the partition. Cluster-marker rankings use the same expression data that defined the clusters, so they describe those clusters rather than independently validate them.

Sex and H2. The 19 features meeting the chosen sex-effect thresholds account for 0.44% of the sum of feature variances. Removing them changes the global results modestly, and the selected clusters do not show an FDR-significant sex association. H2 is not supported by this analysis. This is not evidence that sex has no expression effects or that a small number of genes cannot dominate another dataset; the variance contribution and the actual PCA/clustering results matter.

QC and PCA interpretation. N1 and P25 are sex-marker discrepancy flags; P1 and D10 are lower-correlation flags. Their causes remain unverified. Main analyses retain all samples and their recorded labels. Correlation 0.874 is a dataset-specific diagnostic, not evidence that the arrays are noisier than RNA-seq. Loadings may suggest tissue composition or sample quality hypotheses, but these explanations have not been established by independent QC. The matched 50/46-sample PCA pair fixes the feature selection and scaling rule. The nominal PC2–SI association after exclusions has within-PCA $q = 0.204$; it should not be promoted to a significant result after screening multiple PCs/phenotypes.

Age associations, H3, and H4. Pearson testing yields no 5%-FDR features, but this does not describe all age analyses: UNC13C has Spearman $q = 0.000249$ and adjusted-model $q = 0.0428$ in the full sample. With the four flags excluded, PLAG1 becomes the only adjusted-model feature below 5% FDR ($q = 0.0131$), while UNC13C rises to $q = 0.0894$. The identity of the significant feature is therefore sensitive to sample inclusion. The top-100 Pearson list remains explicitly exploratory. Its age-decreasing features are enriched for several overlapping mitochondrial/metabolic terms, a functional theme consistent with part of H3, but not proof of impaired function or causal aging mechanisms. The T2D group is older and the adjusted/subgroup analyses alter the evidence, supporting concern about confounding in H4. They do not determine how much of the association is caused by age, diabetes, adiposity, or other correlated factors. The non-diabetic subset overlaps the full cohort and is not independent replication.

Relationship to Jin et al. SRF and MKL1 expression have nominal negative associations with SI in this dataset. Some selected target genes also have nominal unadjusted T2D–FH- differences.

These checks reuse the original cohort and do not reproduce the paper's SRF-target gene-set analysis or its animal/cell perturbations. SRF/MKL1 mRNA abundance does not directly measure SRF transcriptional activity. The appropriate claim is partial consistency of expression directions, not independent replication or a verified mechanism. The unadjusted disease comparison is susceptible to age/sex/adiposity confounding; the additional age/sex-adjusted contrast is descriptive and does not remove all such risks.

Limitations and unsuccessful approaches. Weak cluster separation, no Pearson-FDR age features, and sensitivity of the adjusted age signal are substantive results, not reasons to relax a confirmatory threshold. The rank-based top-100 list is a separately labeled exploratory analysis. Small groups, a cross-sectional age range of 21–67 years, age–diabetes imbalance, missing clinical measures, ambiguous gene annotations, and unverified QC flags limit interpretation. No causal effects or general aging biomarker claims are established. A stronger follow-up could use explicitly justified covariates, moderated microarray models such as limma, ranked gene-set tests, and independent data.

10 AI usage and reproducibility

Claude (Anthropic) assisted with the initial analysis code and narrative, including the stored DAVID website queries. Codex (OpenAI) subsequently assisted with statistical review, missing-value and multiplicity corrections, sample-sensitivity analyses, interpretation, and validation. These are AI-assisted contributions; the author is responsible for the final review and submission. This statement does not assert independent human verification of every AI-generated interpretation.

The earlier report was run in GitHub Codespaces/Actions with rocker/verse. This revision was tested on the Boston University SCC through VS Code using R 4.4.3, Pandoc 2.5, and TeX Live 2023. Full session information for the present render follows. Saved DAVID results are imported with an input-membership guard; they are not automatically regenerated. Run the entire document in a fresh R process before submission. Individual chunks share state in the interactive console and can otherwise use stale objects or RNG state.

11 References

1. Jin W, Goldfine AB, Boes T, et al. Increased SRF transcriptional activity in human and mouse skeletal muscle is a signature of insulin resistance. *J Clin Invest.* 2011;121(3):918-929. GEO: GSE25462.
2. López-Otín C, Blasco MA, Partridge L, Serrano M, Kroemer G. The hallmarks of aging. *Cell.* 2013;153(6):1194-1217.
3. Lu T, Aron L, Zullo J, et al. REST and stress resistance in ageing and Alzheimer's disease. *Nature.* 2014;507:448-454 (GDS5204, extra-credit brain dataset, not analyzed here).
4. GEO, GSE25462 and GPL570 sample metadata, accessed October 22, 2026: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE25462.
5. DAVID Functional Annotation Tool documentation, multiple-testing definitions and enrichment background, accessed October 22, 2026: https://davidbioinformatics.nih.gov/documentation/tools/tech-gene-enrichment.html.
6. R `cor.test` documentation (Spearman asymptotic approximation): https://stat.ethz.ch/R-manual/R-devel/library/stats/html/cor.test.html.


```
## R version 4.4.1 (2024-06-14)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.5 LTS
```

```
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;
↪  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] cluster_2.1.8.3 ggrepel_0.9.8   ggplot2_4.0.3   pheatmap_1.0.13
##
## loaded via a namespace (and not attached):
##  [1] Matrix_1.7-0       gtable_0.3.6       dplyr_1.1.4        compiler_4.4.1
##  [5] tidyselect_1.2.1   Rcpp_1.0.13        tinytex_0.53       splines_4.4.1
##  [9] scales_1.4.0       yaml_2.3.10        fastmap_1.2.0      lattice_0.22-6
## [13] R6_2.5.1           labeling_0.4.3     generics_0.1.3     knitr_1.48
## [17] tibble_3.2.1       pillar_1.9.0       RColorBrewer_1.1-3 rlang_1.3.0
## [21] utf8_1.2.4         xfun_0.48          S7_0.2.2           viridisLite_0.4.2
## [25] cli_3.6.3          withr_3.0.2        magrittr_2.0.3     mgcv_1.9-1
## [29] digest_0.6.37      grid_4.4.1         nlme_3.1-164       lifecycle_1.0.4
## [33] vctrs_0.6.5        evaluate_1.0.1     glue_1.8.0         farver_2.1.2
## [37] fansi_1.0.6        rmarkdown_2.28     tools_4.4.1        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```